FILTER DESIGN TEST CASE

High-Pass Filter - n79 (4.4-5.0 GHz 5G NR) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting n79 (4.4-5.0 GHz 5G NR) for GPS/GNSS Anti-Jamming Filter. The filter is designed on AlN on SiC substrate with a target center frequency of 891.1 MHz and bandwidth of 129.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: n79 (4.4-5.0 GHz 5G NR)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: AlN on SiC
- Package: SiP (System in Package)
- Design Tool: Ansys HFSS 2024R1
- Design Center: Newbury Park Design Center

This specification applies to all variants of the High-Pass Filter design targeting n79 (4.4-5.0 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 891.1 MHz
Bandwidth (3 dB): 129.9 MHz
Insertion Loss Target: <= 2.0 dB
Return Loss Target: >= 18.0 dB
Out-of-Band Rejection: >= 36.0 dB
Isolation (Duplexer): >= 42.0 dB
Q Factor: 9902
Temperature Coefficient:-24.2 ppm/°C
Die Size: 2.0 x 0.7 mm
Wafer Size: 6-inch
Number of Resonators: 9
Electrode Material: Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Ansys HFSS 2024R1 and load the High-Pass Filter template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 891.1 MHz, BW 129.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.0 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on SiC).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 2673 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.0 dB
- Return loss in passband shall be >= 18.0 dB
- Out-of-band rejection at +/- 195 MHz offset >= 36.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 2.7 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 86%

6. TEST RESULTS

Measurement Date: 2024-03-31
Devices Tested: 92
Insertion Loss (meas): 1.68 dB
Return Loss (meas): 20.3 dB
Rejection (meas): 39.2 dB
Isolation (meas): 47.1 dB
Q Factor (meas): 9902
Group Delay Variation: 3.1 ns
Power Handling: +28 dBm
Die Yield: 94.7%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The High-Pass Filter design demonstrated below-target performance for n79 (4.4-5.0 GHz 5G NR).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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